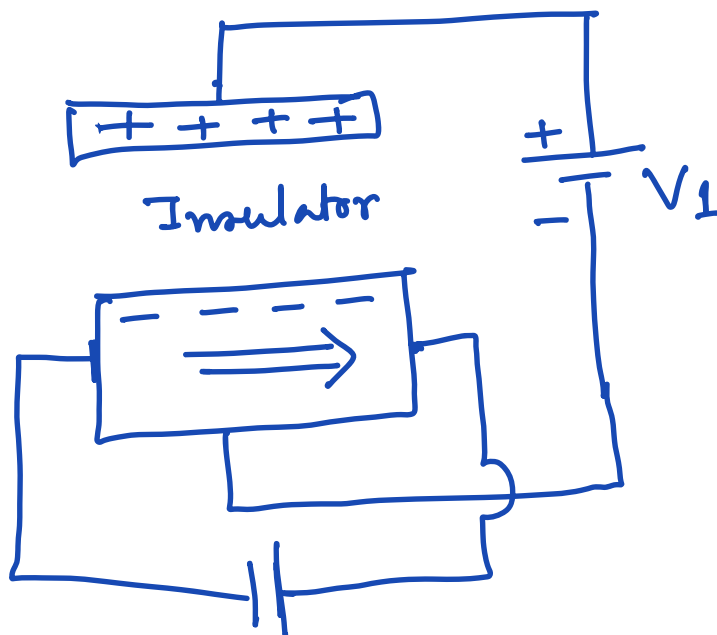
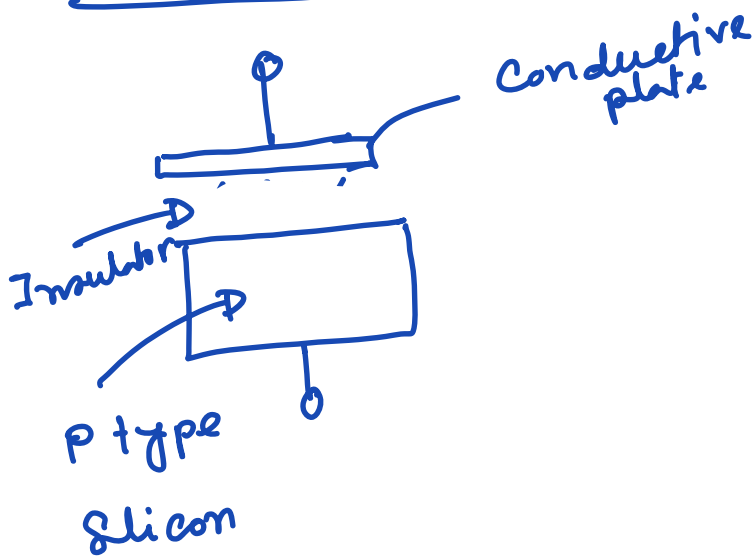


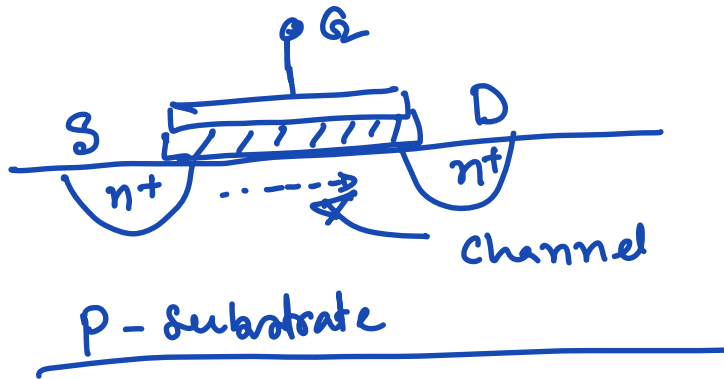
MOSFET

Metal-Oxide-Semiconductor
field effect transistor (MOSFET)

Concept of MOS:



Structure of MOSFET and Symbol



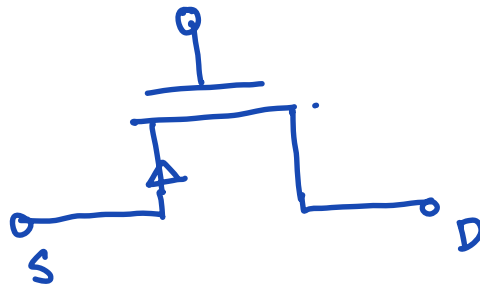
S $\equiv$ Source

D $\equiv$ Drain

G = Gate



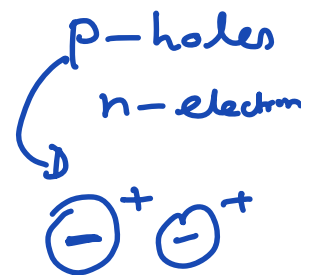
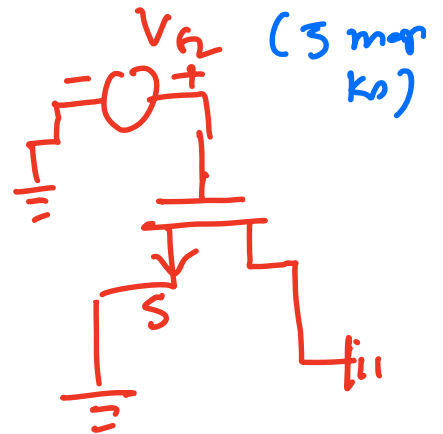
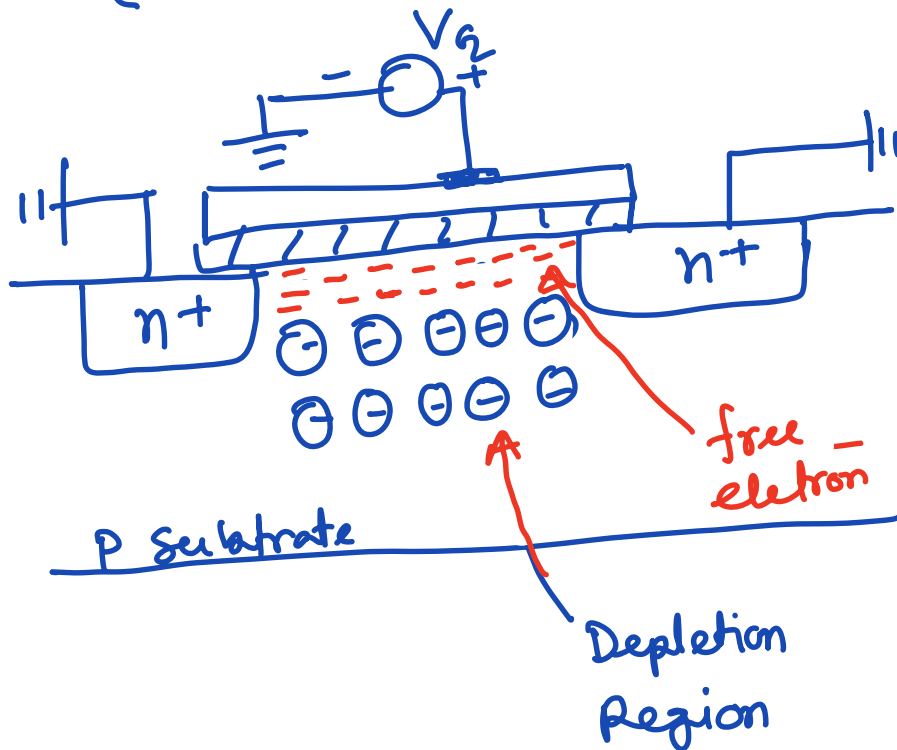
N-MOS



P-MOS

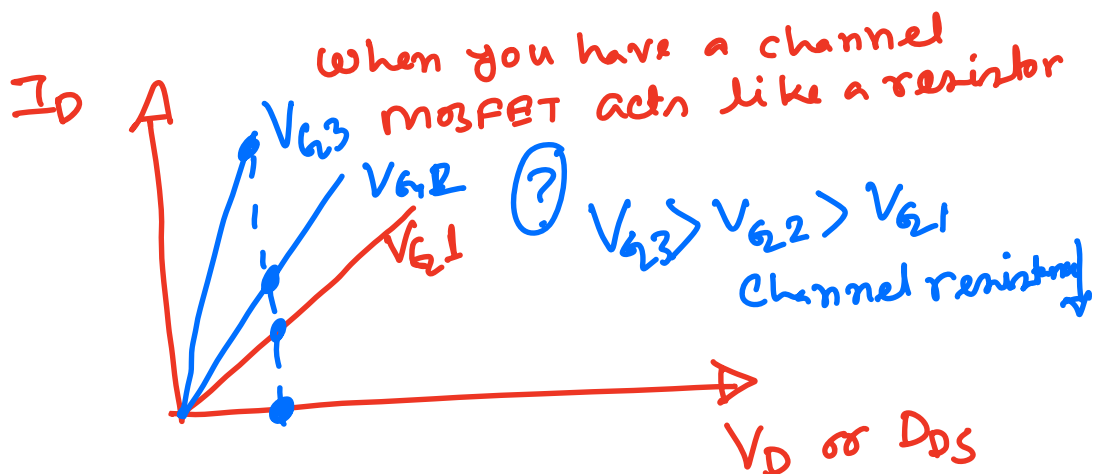
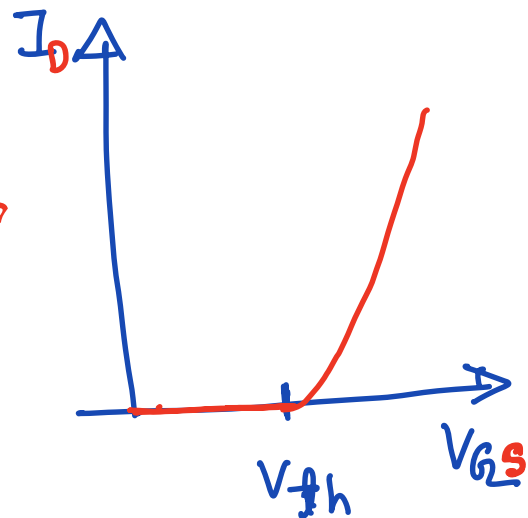
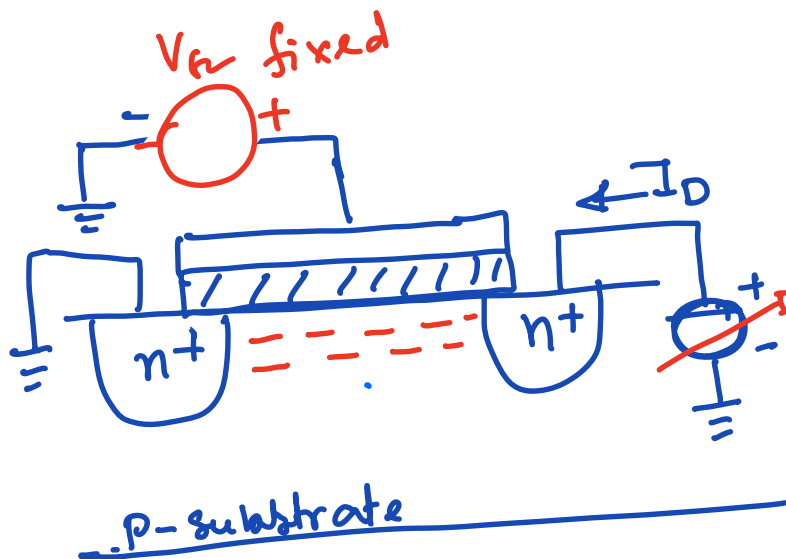
Operation of MOSFET:

(Discuss the operation of MOSFET)



$V_G \geq V_{th}$ then
we will get a channel due
to "inversion"

I-V characteristics :



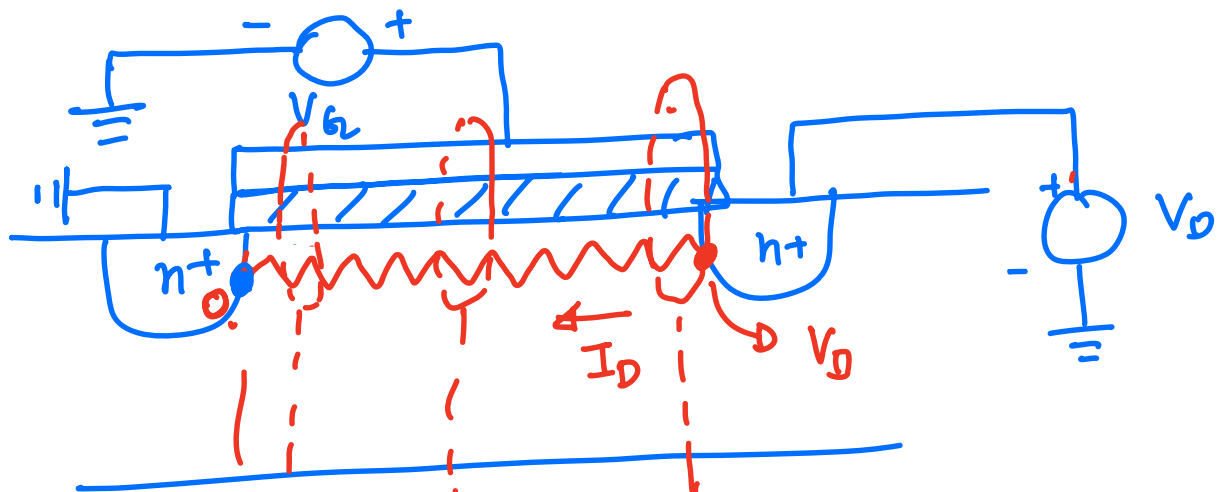
Channel pinch-off :

(Explain concept of channel pinch-off for MOSFET device) (3 marks)

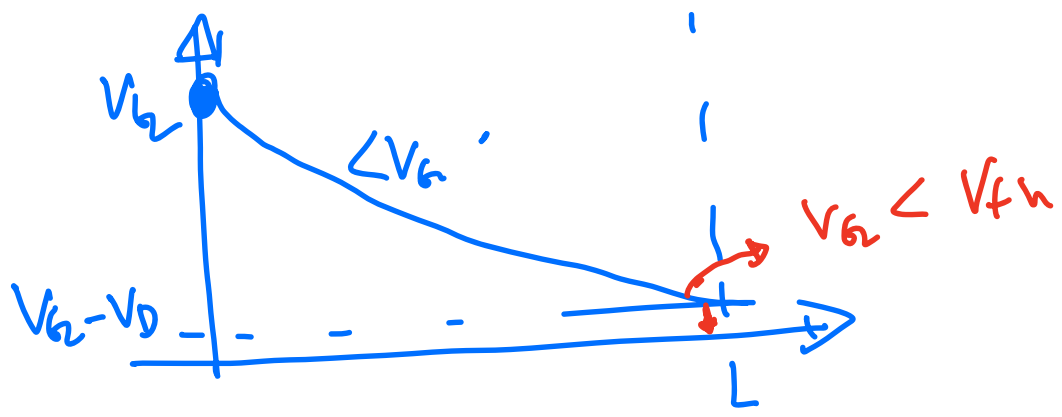
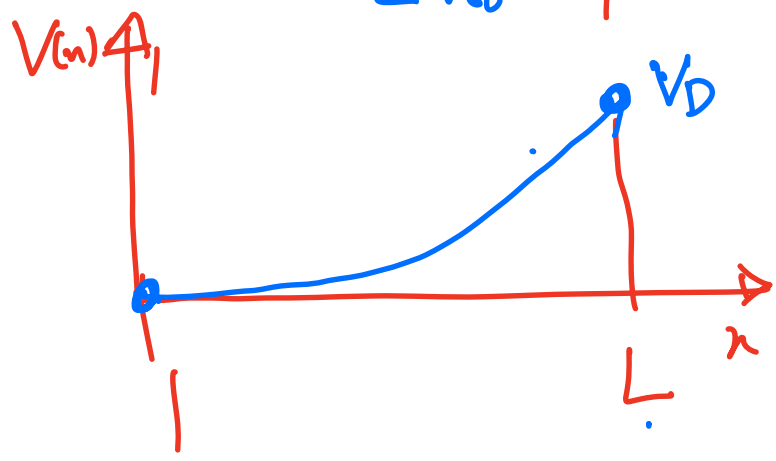
So far what we have learned

① To form a channel gate voltage, V_G must exceed V_{th}

② if the drain voltage remains higher than the source voltage, then the voltage at each point along the channel increases as go from the source to drain.



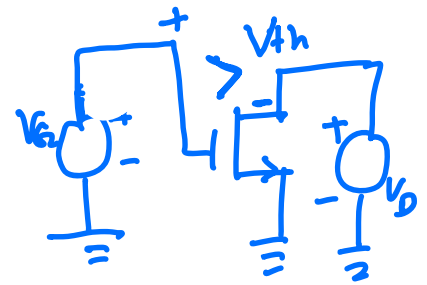
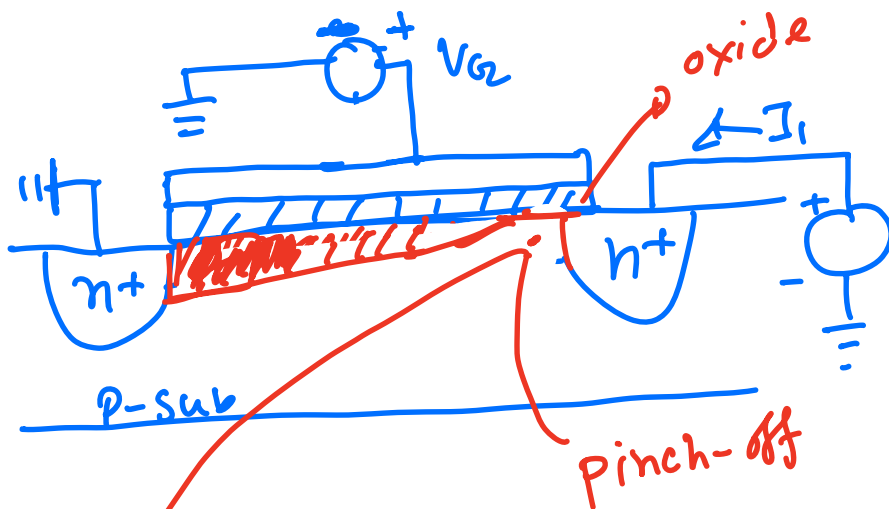
potential difference V_G
 V_G
 $V_G - V_D$
 $< V_G$



Pinch-off:

{ Voltage diff near the source
= V_G
near the drain = $V_G - V_D$
within the channel
 $< V_G$
to form a channel $V_G \geq V_{th}$

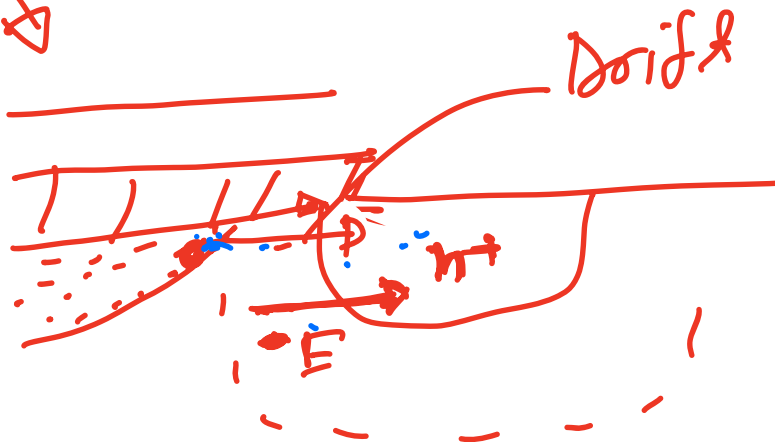
Can we explain channel pinch-off
now?



$$Q_{ch} = C V_G$$

$$V_G - V_D > V_{th}$$

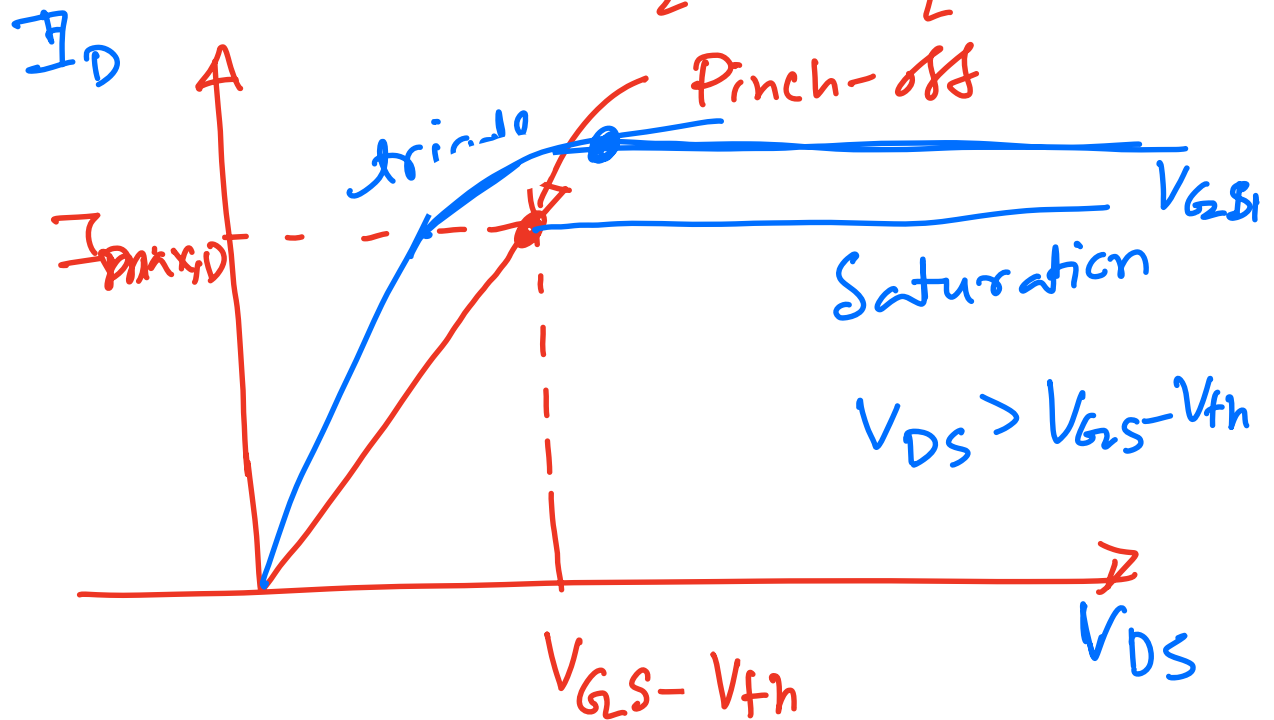
$$V_G - V_{th} > V_D$$



MOSFET IV- Characteristic.

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \left[2(V_{GS} - V_{th}) V_{DS} - V_{DS}^2 \right]$$

$$I_{max,D} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$$



Example .6.3

Triode:

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \left[2 (V_{GS} - V_{th}) V_{DS} - V_{DS}^2 \right]$$

Saturation region

$$I_D = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$$

Overdrive
Voltage

Example 6.6, 6.7